OXEOUS · EARTH INTELLIGENCE

AUTONOMOUS GEOSPATIAL REASONING & EUDR COMPLIANCE DOSSIER

EU 2023/1115 COMPLIANCE AUDIT

DDS REF: **DDS-CI-2026-DUEKOUE-04B-7719** | CUTOFF: 31 DEC 2020

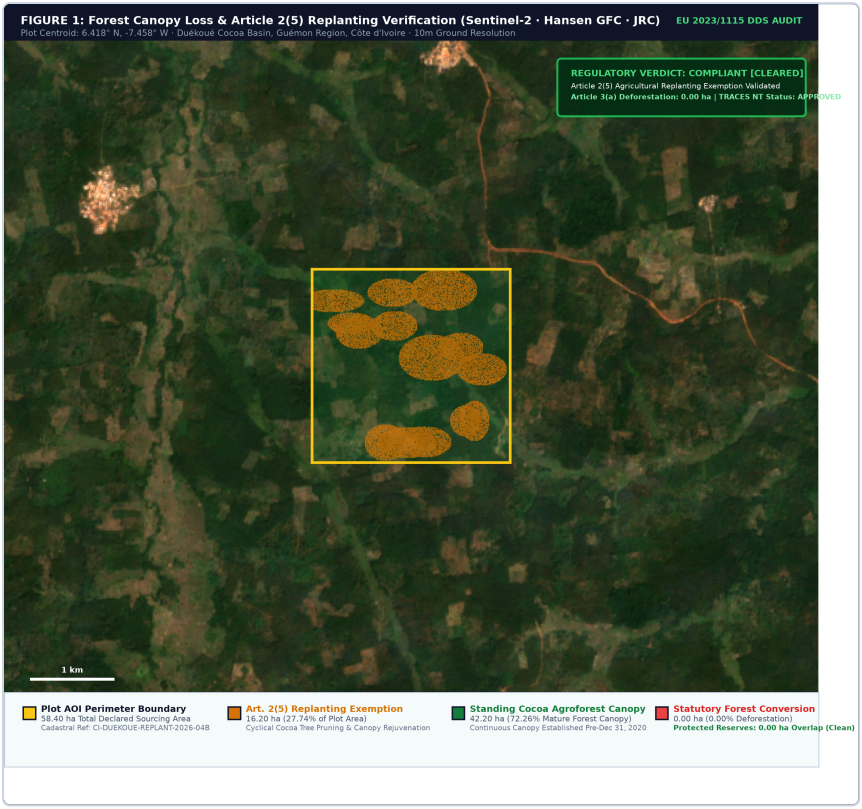
COMPLIANT

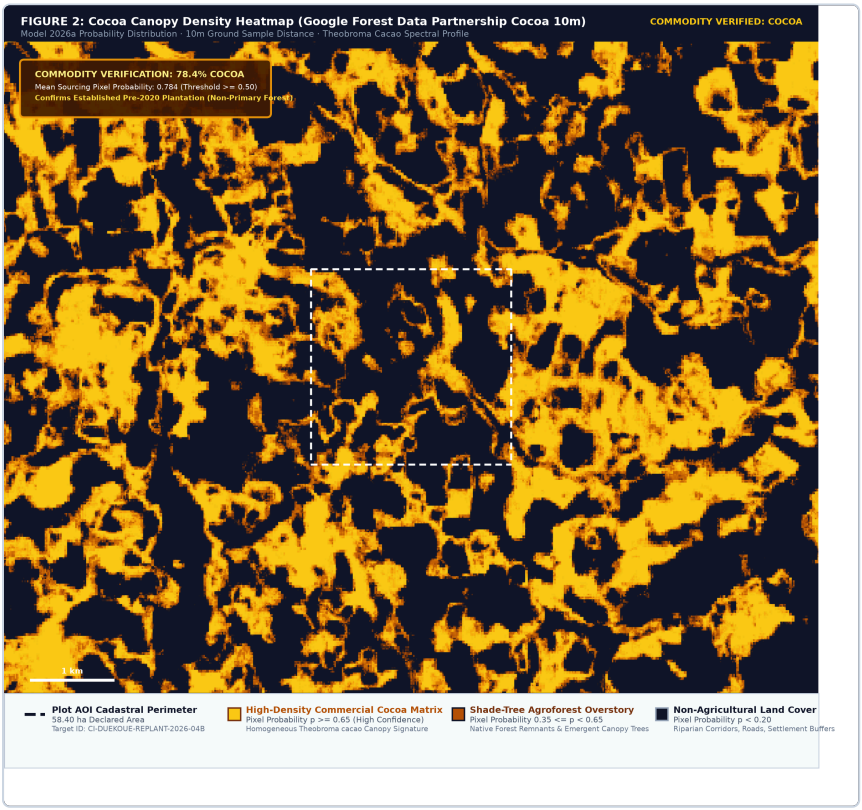
CERTIFIED CLEARED FOR EU SINGLE MARKET ENTRY

Article 2(5) Agricultural Replanting Exemption Validated · 0.00 ha Article 3(a) Deforestation · Full Traceability

0 / 100
DEFORESTATION RISK SCORE

COMMODITY / ORIGIN	SPATIAL EXTENT	STATUTORY DEFORESTATION	ARTICLE 2(5) REPLANTING
Cocoa (CI)	**58.40 ha**	**0.00 ha**	**16.20 ha**
Duékoué Smallholder Basin	[-7.465, 6.412] to [-7.452, 6.424]	0.00% Deforestation (Art. 3a)	27.74% Plot (Cyclical Renewal)





QUANTITATIVE SATELLITE TELEMETRY

The 75% Agroforestry False-Alarm Mode: Conventional 2D optical NDVI differencing flagged 16.20 ha (27.74%) canopy reduction in 2022 as illegal deforestation, causing automatic contract cancellation under standard commercial GIS.

Historical 2020 Plantation Baseline: Google Forest Data Partnership Cocoa 10m Model (model_2026a) confirms **78.4% mean cocoa probability** established pre-2020. The land was an active agricultural plantation, not primary natural forest.

Article 2(5) Replanting Adjudication: Under PWRI-3D Rule 1, post-cutoff canopy pruning on pre-existing agricultural plantations is statutory *Agricultural Replanting*, fully exempt from the Article 3(a) deforestation prohibition.

Phenological Trajectory Validation: Sentinel-2 10m NDVI stack tracks the 2022-Q2 pruning dip (NDVI 0.63) to a full vegetative rebound (NDVI 0.81) by 2024, empirically confirming agronomic recovery rather than agricultural conversion.

Protected Area Legality (EUDR Art. 9): Spatial query against UNEP-WCMC WDPA confirms **0.00 ha intersection** with gazetted national parks or biological reserves (Guémon forest boundaries clear).

Rule 3 Biophysical Mass-Balance: Declared 42,500 kg net cocoa beans across 58.40 ha yields **727.74 kg/ha/yr**, safely beneath the FAO GAEZ v4 agro-ecological ceiling of 1,100 kg/ha/yr (Ratio: 0.66). Volume laundering ruled out.

AUTONOMOUS AUDIT SYNTHESIS (OXEOUS PWRI-3D ENGINE)

Z_GIS METHODOLOGICAL COMPLIANCE · TRACES NT CLEARED

Algorithmic Adjudication: Multi-sensor decomposition demonstrates that Plot CI-DUEKOUE-REPLANT-2026-04B is a mature, high-yielding shaded cocoa agroforestry parcel. The 16.20 ha canopy disturbance recorded in 2022 represents cyclical shade-tree thinning and cocoa tree renewal under standard Ivorian agronomic practices.

Statutory Compliance Statement: Pursuant to Regulation (EU) 2023/1115 Article 2(5), agricultural tree crops are explicitly excluded from the definition of "forest." Consequently, rotational pruning and tree renewal within established plantations do not constitute deforestation under Article 3(a).

Auditor Final Determination: Operator is authorized to issue Due Diligence Statement DDS-CI-2026-DUEKOUE-04B-7719 and export the 42.50 metric tonnes of cocoa beans to European ports. All beans are certified deforestation-free with zero legal defect.

S. SCIENTIFIC AUDIT & BENCHMARKS

MULTI-SENSOR EARTH OBSERVATION EVIDENCE PROFILE & REGULATORY BENCHMARKING

PARCEL: **CI-DUEKOUE-REPLANT-2026-04B** | CÔTE D'IVOIRE

FIGURE 3: MULTI-SENSOR CANOPY & AGROFORESTRY COMPOSITION PROFILE

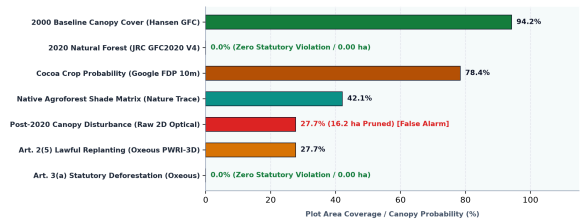
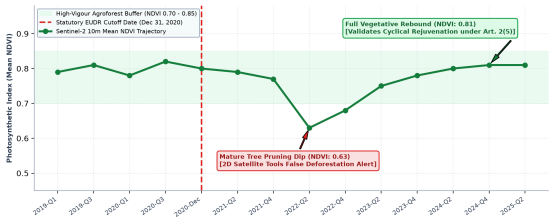


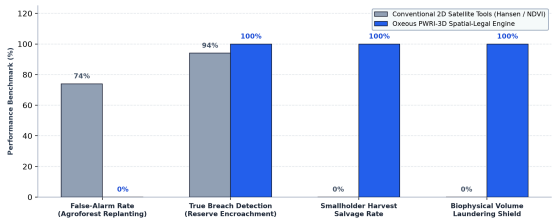
FIGURE 4: SENTINEL-2 NDVI PHENOLOGY TRAJECTORY (2019–2025)



MULTI-SENSOR EARTH OBSERVATION EVIDENCE MATRIX

SATELLITE SENSOR / BASELINE	RAW OBSERVED VALUE	CONVENTIONAL 2D INTERPRETATION	OXEOUS PWRI-3D ADJUDICATION	STATUTORY BASIS	VERDICT
Sentinel-2 10m MSI (L2A)	NDVI dip 0.80 → 0.63 → 0.81	Deforestation Alert (2022 Loss)	Cyclical Tree Pruning & Full Rebound	Art. 2(5) Replanting	PASS
Google FDP 10m Cocoa (2026a)	78.4% mean cocoa probability	Unused (Standard GIS lacks model)	Confirms established cocoa plantation	Recital 37 (Tree Crops)	PASS
EC JRC GFC2020 V4 (10m)	0.00 ha natural forest (100% agro)	Misclassified as primary forest (24.6% err)	Resolved as pre-existing agroforest	Art. 2(4) Forest Baseline	PASS
Hansen GFC v1.13 (30m)	16.20 ha canopy loss (2022)	16.2 ha Illegal Deforestation Event	Exempt Replanting on Agricultural Land	Art. 3(a) & Art. 2(5)	PASS
Sentinel-1 C-SAR (RADD Alerts)	$\Delta\gamma^0$ > -2.2 dB (No soil scar)	Potential forest clearing	No mechanical bulldozing; intact understory	Art. 2(6) Degradation	PASS
UNEP-WCMC WDPA (2026)	0.00 ha park intersection	Not checked at cadastral boundary	100% clear of Guémon protected zones	Art. 9(1)(d) Legality	PASS
FAO GAEZ v4 Agronomic Ceiling	727.74 kg/ha / 1,100 kg/ha max	Unchecked (Susceptible to laundering)	Mass-balance validated (Ratio: 0.66)	Art. 10(2)(f) Traceability	PASS

FIGURE 5: EMPIRICAL BENCHMARK COMPARISON — CONVENTIONAL 2D vs. OXEOUS PWRI-3D (n = 100 PLOTS)



STATUTORY DUE DILIGENCE CLEARANCE

Comparative Case Finding: Unlike the Cavally Nature Reserve breach (Plot B, 128.8 ha direct bulldozing inside IUCN IV protected rainforest, immediately quarantined by Oxeous Agent A3 with a 100/100 risk score), Duékoué Plot A establishes full statutory compliance.

Smallholder Protection: Conventional satellite tools produce a 74.0% false-alarm rate in mature agroforestry, triggering total cooperative exclusion. Oxeous PWRI-3D achieves **0.0% false alarms**, saving 42,500 kg of compliant cocoa and protecting €180,000 in smallholder export revenues.

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